Group -I crossection derivation

Corrected copy. Removed text is in red, new text in blue. The reasons and the numerical checks are in `Group_I_check.pdf`; G1–G6 refer to its Sec. 1.

1 Group I (with c.c.)

1.1 Row I [Done ✓]

Only 4 ρ's

1.2 Row II [Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\left[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\right]\times \left[U^{ab}(w)\mathbb{A}_i^{(3)\,b}(k^+,-w) - \bar{\mathbb{A}}_i^{(3)\,a}(k^+,-w)\right]$$

We write it in two parts:

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x',} e^{-ik(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\,\mu_{\overline{MS}}^2}{4}\right]\right]\times \left[\frac{11}{3}+\log\left[\frac{\Lambda}{k^+}\right]-\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}-1\right]\right]-\frac{67}{9}+\frac{\pi^2}{3}+\log\left[1-\frac{k^+}{\vee}\right]^2\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\right]\times \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^b(x) - U^{ab}(x)\,\rho^b(x)\right] \tag{1.1}$$

Rewwriting, we get

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x',} e^{-ik(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\,\mu_{\overline{MS}}^2}{4}\right]\right]\times \left[\frac{11}{3}+\log\left[\frac{\vee}{k^+}\right]-2\,\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}-1\right]\right]-\frac{67}{9}+\frac{\pi^2}{3}+\log\left[1-\frac{k^+}{\vee}\right]^2\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\right]\times \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^b(x) - U^{ab}(x)\,\rho^b(x)\right] \tag{1.2}$$

Exact bracket (G5a; no limit $\Lambda\to 0$). For $\Lambda < k^+ < \vee - \Lambda$, with $\delta = \Lambda/k^+$ and $P_\epsilon = -\frac{2}{\epsilon} - 2\gamma - \log\frac{(x-w)^2\mu^2}{4}$, the big bracket above is exactly

$$P_\epsilon\Big[\frac{11}{3}+\log\frac{\Lambda^2}{k^{+2}-\Lambda^2}+\log\Big(1-\frac{k^+}{\vee}\Big)-8\delta+2\delta^2-\frac{4}{3}\delta^3\Big]-\frac{67}{9}+\frac{\pi^2}{3}+\log^2\Big(1-\frac{k^+}{\vee}\Big)+\log\Big(1-\frac{k^+}{\vee}\Big)\log\frac{\vee}{k^+}+R_\delta,$$

$$R_\delta = \frac{46}{3}\delta-\frac{4}{3}\delta^2+\frac{8}{9}\delta^3+\Big(\frac{22}{3}-8\delta+2\delta^2-\frac{4}{3}\delta^3\Big)\log(1-\delta)-\Big(8\delta-2\delta^2+\frac{4}{3}\delta^3\Big)\log\delta-2\log\delta\log(1-\delta) - 4\,\mathrm{Li}_2(\delta)-\log^2(1+\delta)+\log(1+\delta)\log\delta.$$

The bracket above is its $\delta\to 0$ limit. $P_\epsilon = \frac{1}{\pi}\int_z\left[K(x,w,z)-K(w,w,z)\right]$ exactly in $d_\perp = 2-\epsilon$, $\overline{\rm MS}$: the same scheme as Row V, part 2b.

$$\frac{d^3N_{2\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4\,f^{abc}}{64\pi^6}\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y,x'} e^{ik\cdot(w-y)}e^{ip\cdot(y-x)}\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}$$
$$\Bigg[\Big[-\log\Big(\frac{\vee}{\Lambda}\Big)+\log\Big[\frac{\vee}{k^+}\Big]\Big]\Big[\frac{k^2p^i}{(k-p)^2}-\frac{k\cdot p\,k^i}{(k-p)^2}\Big]-2\,k^i\,\log\Big(\frac{\vee}{\Lambda}\Big)-2k^i\,\log\Big(\frac{k^+p^2}{\vee\,k^2+k^+p^2}\Big)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}$$
$$\log\Big[1+\Big(\frac{\vee}{k^+}-1\Big)\frac{k^2}{(k-p)^2}\Big]-\frac{(p^2\,k^i+k^2\,p^i)}{(k-p)^2}\Big[\log\Big[\frac{\vee}{\Lambda}\Big]+\log\Big[1-\frac{k^+}{\vee}\Big]\Big]-k^i\,\log\Big[\frac{\vee}{k^+}\Big]\Bigg]\times$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)U^{ah}(w)\,f^{hbc}\,\{\rho^b(\mathbf{x}),\,\rho^c(\mathbf{y})\}-f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(\mathbf{x}),\,\rho^e(\mathbf{y})\}\Big] \tag{1.3}$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{d^3N_{2\,a}}{d^3k} + \frac{d^3N_{2\,b}}{d^3k} \tag{1.4}$$

$$\begin{aligned}\frac{d^3N_{2\,a}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{i\,g^4}{32\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{\vee}{\Lambda}\right) - \log\left(\frac{\vee}{k^+}\right)\right]\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x'}\frac{(x'-w')^i}{(x'-w')^2}\,\mathfrak{F}^i(w,x,y)\\ &\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[U^{ah}(w)\,f^{hbc}\left\{\rho^b(x),\rho^c(y)\right\} - f^{abc}\,U^{bd}(x)\,U^{ce}(y)\left\{\rho^d(x),\rho^e(y)\right\}\Big]\end{aligned} \tag{1.5}$$

with

$$\begin{aligned}\mathfrak{F}^i(w,x,y) &= \frac{(x-w)^i}{(x-w)^2}\int_z\frac{(w-z)\cdot(y-z)}{(w-z)^2(y-z)^2} - \frac{(y-w)^i}{(y-w)^2}\int_z\frac{(w-z)\cdot(x-z)}{(w-z)^2(x-z)^2}\\ &\quad - \left[\frac{(x-w)^i}{(x-w)^2} - \frac{(y-w)^i}{(y-w)^2}\right]\int_z\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}\end{aligned} \tag{1.6}$$

$$= -\pi\left[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2} - \frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\right]. \tag{1.7}$$

Operator ordering (G3). `Calculation_main.tex` drops $-[\mathbb{N}^{(2)},\mathbb{A}_2^{(1)}]$ in the unitarity relation (its correction E5). With the virtual factor ordered in p^+, part 2a is

$$\begin{aligned}\frac{d^3N_{2\,a}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{i\,g^4}{32\pi^5}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x'}\frac{(x'-w')^i}{(x'-w')^2}\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\rho^{b'}(x')\\ &\times\Big\{\Big[\log\left(\frac{\vee}{\Lambda}\right)\mathfrak{F}^i_\beta - \log\left(\frac{\vee}{k^+}\right)\mathfrak{F}^i\Big]U^{ah}(w)f^{hbc}\{\rho^b(x),\rho^c(y)\} - \Big[\log\left(\frac{\vee}{\Lambda}\right) - \log\left(\frac{\vee}{k^+}\right)\Big]\mathfrak{F}^i f^{abc}U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\}\Big\},\end{aligned}$$

where $\mathfrak{F}^i_\beta$ is the first line of $\mathfrak{F}^i$ (the two terms with $\int_z\frac{(w-z)\cdot(\cdots)}{\cdots}$). Only the three-$\rho$ term $\propto\log(\vee/\Lambda)$ of the $U^{ah}(w)$ part changes.

$$\begin{aligned}\frac{d^3N_{2\,b}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4\,f^{abc}}{64\pi^6}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y,x'} e^{ik\cdot(w-y)}\,e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,R^i(\mathbf{k},\mathbf{p})\\ &\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[U^{ab}(w)U^{ah}(w)\,f^{hbc}\left\{\rho^b(x),\rho^c(y)\right\} - f^{abc}\,U^{bd}(x)\,U^{ce}(y)\left\{\rho^d(x),\rho^e(y)\right\}\Big]\end{aligned} \tag{1.8}$$

where

$$R^i = -2k^i\log\left(\frac{p^2}{k^2+\frac{k^+}{\vee}p^2}\right) + \frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\,\log\left[\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee}\left(1-\frac{k^2}{(k-p)^2}\right)\right] - \frac{p^2k^i+k^2p^i}{(k-p)^2}\,\log\left(1-\frac{k^+}{\vee}\right). \tag{1.9}$$

Exact R^i (G5b). With $E^i=\frac{k^2p^i-k\cdot p\,k^i}{(k-p)^2}$, $C^i=\frac{(2p^2-2k\cdot p+k^2)k^i+k^2p^i}{(k-p)^2}$, $D^i=\frac{p^2k^i+k^2p^i}{(k-p)^2}$ and $\Lambda<k^+<\vee-\Lambda$:

$$\begin{aligned}R^i_{\mathrm{exact}} = &-2k^i\log\frac{p^2+\frac{\Lambda}{k^+}k^2}{k^2+\frac{k^+}{\vee}p^2} + C^i\log\left[\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee}\left(1-\frac{k^2}{(k-p)^2}\right)\right] - D^i\log\left(1-\frac{k^+}{\vee}\right)\\ &+E^i\log\frac{k^+}{k^+-\Lambda} - C^i\log\left[1+\frac{\Lambda}{k^+}\frac{k^2}{(k-p)^2}\right] + k^i\log\left(1+\frac{\Lambda}{k^+}\right).\end{aligned}$$

Part 2a does not change.

1.3 Row III

[Done ✓]

$$\begin{aligned}\frac{d^3N_{\mathrm{NLO}}}{d^3k} = &\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int\frac{dp^+}{2\pi}\int_z\Big[-\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w') + \mathbb{A}^{\dagger\,(1)\,b'}_i(k^+,-w')U^{b'a}(w')\Big]\times\\ &\Big[-\bar{\mathbb{B}}^{\dagger\,ba}_{3ji}(k^+,w;p^+,-z)\bar{\mathbb{A}}^{(1)\,b}_j(p^+,-z) + \bar{\mathbb{B}}^{\dagger\,ba}_{3ji}(k^+,w;p^+,-z)U^{bc}(z)\mathbb{A}^{(1)\,c}_j(p^+,-z)\Big]\end{aligned}$$

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ & \\
& \frac{ig^4\, f^{abd}}{8\pi^4}\int_{k^+ +\Lambda}^{\vee}\frac{dp^+}{2\pi}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(p^+ + k^+)}{(k^+ - p^+)^2}\,\frac{1}{[p^+\,(x-z)^2 - k^+\,(x-w)^2]} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& + \frac{ig^4\, f^{abd}}{8\pi^4}\int_{\Lambda}^{k^+ -\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(p^+ + k^+)}{(k^+ - p^+)^2}\,\frac{1}{[p^+\,(x-z)^2 - k^+\,(x-w)^2]} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& - \frac{ig^4\, f^{abd}}{8\pi^4}\int_{\Lambda}^{\vee}\frac{dp^+}{2\pi}\,\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\frac{[k^+\,(x-w)^2 + p^+\,(x-z)^2]}{[p^+\,(x-z)^2 - k^+\,(x-w)^2]}\,\frac{1}{p^+\,k^+} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& + - \frac{ig^4\, f^{abd}}{4\pi^4}\,\frac{1}{k^+}\int_{\Lambda+k^+}^{\vee}\frac{dp^+}{2\pi}\int_{x,y,z,x'}\frac{(x'-w')^i\,(y-z)^j\,(z-w)^k}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\,\frac{[k^+\,(x-w)^m - p^+\,(x-z)^m]}{[k^+\,(x-w)^2 - p^+\,(x-z)^2]}\,\frac{1}{(k^+ - p^+)} \\
& \left[\delta_{ik}\,\delta_{jm} - \frac{k^+}{p^+}\,\delta_{jk}\,\delta_{im} - \frac{k^+}{(k^+ - p^+)}\,\delta_{ij}\,\delta_{km}\right]\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times \\
& \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& - \frac{ig^4\, f^{abd}}{4\pi^4}\int_{\Lambda}^{k^+ -\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'}\frac{(x'-w')^i\,(y-z)^j\,(z-w)^k}{(x'-w')^2\,(y-z)^2\,[k^+\,(x-w) - p^+\,(x-z)]^2}\,\frac{[k^+\,(x-w)^m - p^+\,(x-z)^m]}{[k^+\,(x-w)^2 - p^+\,(x-z)^2]}\,\frac{p^+}{(k^+ - p^+)} \\
& \left[\delta_{ik}\,\delta_{jm} - \frac{k^+}{p^+}\,\delta_{jk}\,\delta_{im} - \frac{k^+}{(k^+ - p^+)}\,\delta_{ij}\,\delta_{km}\right]\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times \\
& \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\}
\end{aligned}
\tag{1.10}
$$

Sign (G1). The non-instantaneous entry of $\mathbb{B}_3$ for $p^+ > k^+ + \Lambda$ needs a minus sign (correction E3 of `Calculation_main.tex`). Hence the sign of the fourth line above, and of every line of $d^3N_{2\,\rm NLO}$ below.

Integrating wrt p^+ and we write it in three parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ & \\
& \frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[(x-z)^2-(x-w)^2]}\times\left[\frac{1}{\Lambda}+\frac{1}{k^+-\vee}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{\left[\vee\,(x-z)^2-k^+\,(x-w)^2\right]}{\left[(\Lambda+k^+)\,(x-z)^2-k^+\,(x-w)^2\right]}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[(x-z)^2-(x-w)^2]}\times\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{k^+-\Lambda}\right]-\log\left[\frac{\vee}{\Lambda}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{\left[(k^+-\Lambda)\,(x-z)^2-k^+\,(x-w)^2\right]}{\left[\Lambda\,(x-z)^2-k^+\,(x-w)^2\right]}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{\vee}{\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left[\frac{\left[k^+\,(x-w)^2-\vee\,(x-z)^2\right]}{\left[k^+\,(x-w)^2-\Lambda\,(x-z)^2\right]}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \qquad (1.11)
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,[(x-w)^2-(x-z)^2]}\left[\frac{1}{(\vee-k^+)}-\frac{1}{\Lambda}\right] \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda+k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right]\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right]\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\Bigg\}
\end{aligned}
\tag{1.12}
$$

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& \frac{ig^4 f^{abd}}{8\pi^5}\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\mathcal{A}\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\mathcal{A}\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\mathcal{A}\int_{x,y,z,x'}\frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\Bigg\}
\end{aligned}
\tag{1.13}
$$

$$
\begin{aligned}
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
& +\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,(z-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\}
\end{aligned}
\tag{1.14}
$$

where

$$
\begin{aligned}
\mathcal{A} = \frac{1}{\sqrt{(x-w)^2(x-z)^2-\left[(x-w)\cdot(x-z)\right]^2}}\Bigg\{ & \arctan\left[\frac{(k^+-\Lambda)(x-z)^2-k^+(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2(x-z)^2-\left[(x-w)\cdot(x-z)\right]^2}}\right] \\
& -\arctan\left[\frac{\Lambda(x-z)^2-k^+(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2(x-z)^2-\left[(x-w)\cdot(x-z)\right]^2}}\right]\Bigg\}
\end{aligned}
$$

Principal values (G4). The integrand has a pole at $p^+_* = k^+(x-w)^2/(x-z)^2$. When it lies inside the range, the p^+ integral is a principal value, and every logarithm of a ratio of $k^+(x-w)^2 - c\,(x-z)^2$ factors in d^3N_1, d^3N_2, d^3N_{3a} must be read as $\log|\cdots|$. **Green terms.** With the sign of d^3N_2 corrected, the four green terms cancel exactly: $\left[\frac{1}{\Lambda}+\frac{1}{k^+-\vee}\right]+\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right]-\left[\frac{1}{\Lambda}-\frac{1}{\vee-k^+}\right]-\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right]=0$, all multiplying $\frac{ig^4 f^{abd}}{8\pi^5}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2[(x-z)^2-(x-w)^2]}$ and the same colour factor. As written they leave $2\left[\frac{1}{\Lambda}-\frac{1}{\vee-k^+}\right]$.

1.4 Row IV

[Done ✓]

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{2}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int_{\Lambda}^{\vee-k^+}\frac{dp^+}{2\pi}\int_z & \Big[-\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w') + \mathbb{A}^{\dagger\,(1)\,b'}_i(k^+,-w')U^{b'a}(w')\Big]\times \\
& \Big[U^{ac}(w)U^{bd}(z)\bar{\mathbb{A}}^{\dagger\,(1)\,b}_j(p^+,-z)\mathbb{B}^{dc}_{2ji}(k^+,-w;p^+,-z) - \bar{\mathbb{A}}^{\dagger\,(1)\,b}_j(p^+,-z)\bar{\mathbb{B}}^{ba}_{2ji}(k^+,-w;\,p^+,-z)\Big]
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{4\pi^4}\int_{\Lambda}^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,z,w,x',w'} & e^{-ik(w'-w)}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[p^+(x-z)^2+k^+(x-w)^2]} \\
& \left[\frac{\delta^{ij}}{2\,(z-w)^2\,(p^++k^+)}\left[(x-z)^2-(x-w)^2\right]+\frac{1}{k^+}\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2}-\frac{(x-z)^j\,(x-w)^i}{2\,(x-w)^2}\right]\right. \\
& \left.+\frac{1}{p^+}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2}\right]\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \\
& \Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\Big]
\end{aligned}
\tag{1.15}
$$

Integrating w.r.t. p^+

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\Bigg\{ \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]\times\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2} - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2} + \frac{\delta^{ij}}{2\,(z-w)^2}\right] - \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\times\log\left[\frac{\vee}{\Lambda+k^+}\right] + \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\left[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\right] \tag{1.16}$$

1.5 Row V

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{2}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int\frac{dp^+}{2\pi}\int_{x,z}\left[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\right]\times \left[-\bar{\mathbb{C}}^{\dagger\,bca}_{1jki}(k^+,w-z;p^+,z-x)\bar{\mathbb{B}}^{bc}_{2jk}(k^+-p^+,-z;p^+,-x) + \bar{\mathbb{C}}^{\dagger\,dca}_{1jki}(k^+,w-z;p^+,z-x)\,U^{db}(x)\,U^{ce}(z)\,\mathbb{B}^{be}_{2jk}(k^+-p^+,-z;p^+,-x)\right]$$

We write it in two parts:

1.5.1 First part: [Done ✓]

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',y'}\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\,\delta^{(2)}\left[w-z+\frac{p^+}{k^+}\,(z-x)\right] \left[\delta_{im}\,\delta_{jk}-\frac{k^+}{p^+}\,\delta_{jm}\,\delta_{ik}-\frac{k^+}{(k^+-p^+)}\,\delta_{ij}\,\delta_{km}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times \left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]\Bigg\} \tag{1.17}$$

Integrating w.r.t. w, we get

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\Bigg\{\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \left[\delta_{im}\,\delta_{jk}-\frac{k^+}{p^+}\,\delta_{jm}\,\delta_{ik}-\frac{k^+}{(k^+-p^+)}\,\delta_{ij}\,\delta_{km}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times \left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]\Bigg\} \tag{1.18}$$

Limits (G5c). The plus prescription is taken on $\Lambda < p^+ < k^+ - \Lambda$: $\int_\Lambda^{k^+-\Lambda} dp^+\,[\frac{k^+}{p^+}]_+ f \equiv \int_\Lambda^{k^+-\Lambda} dp^+\,\frac{k^+}{p^+}[f(p^+)-f(0)]$. Then the endpoint terms, with $\log\frac{k^+-\Lambda}{\Lambda} = \log\frac{\vee}{\Lambda} + \log\frac{k^+-\Lambda}{\vee}$, are exact. With $\int_0^{k^+}$ the result is off by $O(\Lambda/k^+)$.

Using + prescription, we get

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\, \frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_0^{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\left[\delta_{im}\,\delta_{jk} - \left[\frac{k^+}{p^+}\right]_+\delta_{jm}\,\delta_{ik}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times$$
$$\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]$$
$$-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-x)}\, e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dq^+}{2\pi}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\,\left[\frac{1}{q^+}\right]_+$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]$$
$$-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]$$
$$-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-x)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right] \tag{1.19}$$

1.5.2 Second part:

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik\cdot(w'-w)}\left\{\right.$$
$$\frac{g^4}{8\pi^4}\,\frac{1}{k^{+\,3}}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'}\frac{p^+\,(k^+-p^+)}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\,\delta^{(2)}\left[w-z+\frac{p^+}{k^+}\,(z-x)\right]\times$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right] - \frac{k^{+\,2}}{(k^+-p^+)^2}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{k^+\,\delta_{im}}{p^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right] - \frac{k^{+\,2}}{p^{+\,2}}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{k^+\,\delta_{im}}{(k^+-p^+)}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.\left.-\frac{k^{+\,2}}{p^+\,(k^+-p^+)}\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\right.$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[f^{adc}\,f^{ebf}\,U^{db}(x)\,U^{ce}(z) - f^{abc}\,f^{cbd}\,U^{df}(y)\right]\rho^{b'}(x')\,\rho^f(y)\left.\right\} \tag{1.20}$$

Integrating w.r.t. w, we get

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}} \Bigg\{ \frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{\Lambda}^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'}\frac{1}{\left[p^+(y-x)^2+(k^+-p^+)(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+\,(k^+-p^+)}{k^{+\,2}} - \frac{p^+}{(k^+-p^+)}\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\Bigg\}$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[f^{adc}\,f^{ebf}\,U^{db}(x)\,U^{ce}(z)-f^{abc}\,f^{cbd}\,U^{df}(y)\right]\rho^{b'}(x')\,\rho^{f}(y)\Bigg\} \tag{1.21}$$

This can be written as:

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}} \Bigg\{ \frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{\Lambda}^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'}\frac{1}{\left[p^+(y-x)^2+(k^+-p^+)(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+\,(k^+-p^+)}{k^{+\,2}} - \frac{p^+}{(k^+-p^+)}\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\Bigg\}$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c U^{af}(y)\right]\rho^{b'}(x')\,\rho^{f}(y)\Bigg\} \tag{1.22}$$

This term contains a UV divergence at $x \to z$. So, we use the following identity to isolate the UV term and the finite term. The identity as follows:

$$f^{abc}\,f^{def}\,U^{be}(z)\,U^{cf}(z) = N_c\,S^{ad}(z)\,U^{ad}(z)$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}} \Bigg\{$$

$$\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'}\frac{1}{\left[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$

$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}} - \frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$

$$+\,\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$

$$-\,\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$

$$+\,\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$

$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\Bigg\}$$

$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(y)+N_c\,U^{af}(x)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\} \tag{1.23}$$

This can be written in two parts:

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}} \Bigg\{$$

$$\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{1}{\left[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$

$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}} - \frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$

$$+\,\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$

$$-\,\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$

$$+\,\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$

$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\Bigg\}$$

$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\} \tag{1.24}$$

$$
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}} \Bigg\{
$$

$$
\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'}\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}
$$

$$
\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}} - \frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.
$$

$$
+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]
$$

$$
-\frac{(k^+-p^+)}{p^+}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]
$$

$$
+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]
$$

$$
\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\Bigg\}
$$

$$
\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\} \tag{1.25}
$$

Limits (G5c). The plus prescription is taken on $\Lambda < p^+ < k^+ - \Lambda$: $\int_\Lambda^{k^+-\Lambda} dp^+\,[\frac{k^+}{p^+}]_+ f \equiv \int_\Lambda^{k^+-\Lambda} dp^+\,\frac{k^+}{p^+}[f(p^+)-f(0)]$. Then the endpoint terms, with $\log\frac{k^+-\Lambda}{\Lambda} = \log\frac{\vee}{\Lambda} + \log\frac{k^+-\Lambda}{\vee}$, are exact. With $\int_0^{k^+}$ the result is off by $O(\Lambda/k^+)$.

We will handle the p^+, using + prescription for the first part. We get the following:

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_0^{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\,e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\Bigg\{$$
$$\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'}\frac{1}{\left[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}$$
$$+\,\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\left[\frac{k^+}{p^+}\right]_+\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\,\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\Bigg]\Bigg\}$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\}$$
$$+\,\frac{1}{(2\pi)^3}\int_{k^+}^{0}\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}\,e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'}\frac{1}{\left[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\Bigg\{\left[\frac{k^+}{q^+}\right]_+\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\Bigg\}\times$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$-\,\frac{1}{(2\pi)^3}\int_{w'}e^{-ik\cdot(w'-z)}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.$$
$$\left.+\,\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$-\,\frac{1}{(2\pi)^3}\int_{w'}e^{-ik\cdot(w'-x)}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}$$
$$\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{1.26}$$

For the second part, we will first handle the UV divergence and then do plus prescription to handle the p^+.

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k}+\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k}$$

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[2\log\left(\frac{\vee}{\Lambda}\right)-2\log\left(\frac{\vee}{k^+}\right)+2\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{6}+\frac{4\Lambda}{k^+}-\frac{\Lambda^2}{k^{+2}}+\frac{2\Lambda^3}{3k^{+3}}\right]$$
$$\times\int_{y,z,x',w'}e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y). \tag{1.27}$$

Setting $\Lambda\to 0$, we get (G5d) Keep the exact form above: the form below drops $\frac{4\Lambda}{k^+}-\frac{\Lambda^2}{k^{+2}}+\frac{2\Lambda^3}{3k^{+3}}$ but keeps $2\log(1-\frac{\Lambda}{k^+})$, which is of the same order.

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[2\log\left(\frac{\vee}{\Lambda}\right)-2\log\left(\frac{\vee}{k^+}\right)+2\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{6}\right]\int_{y,z,x',w'}e^{-ik\cdot(w'-z)}$$
$$\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y). \tag{1.28}$$

$$\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Bigg\{$$

$$e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$

$$\times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\big[(y-x)^2-(y-z)^2\big]\,\frac{p^+(k^+-p^+)}{k^{+2}}-\frac{p^+}{k^+-p^+}\,\Bigg[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\Bigg]$$

$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\Bigg[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Bigg]$$

$$-\frac{k^+-p^+}{p^+}\,\Bigg[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\Bigg]$$

$$+\frac{p^+\,\delta_{im}}{k^+}\,\Bigg[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Bigg]$$

$$-\Bigg[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\Bigg]\Bigg]$$

$$\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)$$

$$-\frac{1}{k^+}\,\Bigg[\frac{p^+(k^+-p^+)}{k^{+2}}+\frac{p^+}{k^+-p^+}+\frac{k^+-p^+}{p^+}\Bigg]\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}$$

$$\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)\Bigg\}. \tag{1.29}$$

Now, we do plus prescription for $\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k}$.

$$\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} = \tag{1.30}$$